Aakash
Medical | IIT-JEE | Foundations
Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **Project Phoenix for Re-NEET-2026_PST-04P** **Time : 180 Min.**

Topics Covered:
Physics: Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor, Experimental Skills
Chemistry: Solutions, Electrochemistry, Chemical Kinetics, The *p*-Block Elements(Group-15 to 18), The *d* & *f*-Block Elements, Coordination Compounds, Principles Related to Practical Chemistry
Botany: Sexual reproduction in flowering plants, Principles of inheritance and variations, Biodiversity and conservation
Zoology: Reproductive Health, Human Health & Disease, Evolution

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. The cylindrical lens is required to correct
(1) Myopia (2) Hypermetropia
(3) Presbyopia (4) Astigmatism

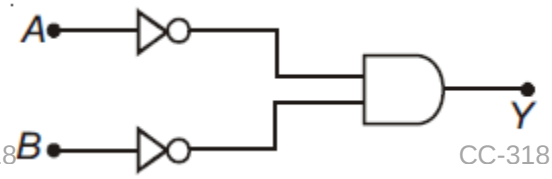
2. A plane mirror forms image of an object, then it will be
(1) Always real
(2) Always virtual
(3) Erect and of same size
(4) Erect and magnified in size

3. The wavelength of the Balmer series of the hydrogen atom mostly appears in
(1) UV region (2) Visible region
(3) IR region (4) Radio wave region

4. Regarding nature of electromagnetic waves **incorrect** statement is
(1) Electric and magnetic fields in electromagnetic wave are perpendicular to each other
(2) Electric field and magnetic field are perpendicular to direction of propagation of electromagnetic wave
(3) Electromagnetic waves has self-sustaining oscillations of electric and magnetic field in vacuum
(4) The velocity of electromagnetic wave is independent of electric and magnetic properties of the medium

5. Consider the following statements:
Statement A: Energy of a photon depends upon its frequency and it does not change with change in medium.
Statement B: Photons are electrically neutral and they are not deflected by electric or magnetic fields.
Statement C: A photon does not exist at rest and its rest mass is zero.
Based on above information pick the correct option.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Only statements A and B are incorrect
(4) All statements are correct

6. What is the output in the figure?



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(1) $\overline{A.B}$ (2) AB
(3) $\overline{A+B}$ (4) $\overline{A+B}$

7. In a nuclear reaction

$$^{A}_{Z}X \rightarrow Y + 2^{0}_{+1}\beta + 2\nu$$

$$Y \rightarrow {}^{229}_{89}Z + 2\alpha + \text{Energy}$$

Determine the mass number and atomic number of element *X*?

(1) 237, 93 (2) 237, 95
(3) 237, 91 (4) 221, 84

8. An electromagnetic wave is expressed by the equation $y(x,t) = 0.05\sin\left(3\times10^{8}t - 15x\right)$ m
(All the given quantities are in SI unit)
The velocity of wave is

(1) 8×10^{7} m/s (2) 0.4×10^{8} m/s
(3) 6×10^{7} m (4) 2×10^{7} m/s

9. What will be the angular width of central maxima in Fraunhofer diffraction when light of wavelength 6000 Å is used and slit width is 12×10^{-5} cm?

(1) 3 rad (2) 8 rad
(3) 1 rad (4) 2 rad

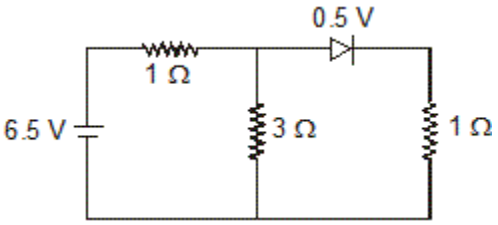
10. In an equilateral prism, if angle of minimum deviation is 30°, then incident angle is

(1) 45° (2) 30°
(3) 90° (4) 0°

11. Two thin prisms A and B are made of glass of refractive index 1.4 and 1.5. They are combined with each other so that this combination produces dispersion without deviation. If refracting angle of prism A is 5° then the refracting angle of second prism will be

(1) 5° (2) 3°
(3) 2° (4) 4°

12. In the circuit shown below, the diode has a forward voltage drop of 0.5 V. What is the value of the power dissipated by diode?



(1) 4.2 watt (2) 1.25 watt
(3) 7 watt (4) 12.5 watt

13. A plano convex lens is silvered on its plane side. The radius of curvature of the lens is 10 cm and the refractive index of the material of lens is 3/2. An object is placed 20 cm in front of the silvered lens, the position of final image is

(1) 10 cm (2) 20 cm
(3) 30 cm (4) Infinity

14. The energy released by fission of one uranium atom is 200 MeV. The number of fissions per second required to produce 6.4 W power will be

(1) 10^{11} (2) 2×10^{11}
(3) 10^{10} (4) 2×10^{10}

15. The number of neutrons present in a nucleus having atomic number 11 and mass number 24 is

(1) 11 (2) 13
(3) 10 (4) 24

16. If the frequency of incident light in a photoelectric experiment is doubled, the stopping potential will

(1) Be doubled
(2) Be halved
(3) Become more than doubled
(4) Become less than doubled

17. The ratio of shortest wavelength of Lyman series to the shortest wavelength of Paschen series is

(1) 9 : 5 (2) 1 : 9
(3) 5 : 36 (4) 4 : 9

18. Given below are two statements:
Statement I: For a concave mirror, the minimum distance between a real object and its real image is 4*f*.
Statement II: A virtual image can be photographed.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

19. The kinetic energy of a particle is reduced to $\frac{1}{4}$th of its initial value keeping its mass fixed, then de-Broglie wavelength of this particle will be

(1) Doubled (2) Halved
(3) Quadrupled (4) Unchanged

20. An equiconvex lens of refractive index 1.5 has power 4 D in air. Its power in water is (R.I. of water = $\frac{4}{3}$)

(1) 2.0 D (2) 1.0 D
(3) 3.0 D (4) 0.75 D

21. The binding energy per nucleon for deuteron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus $_2He^4$ is

(1) 26.3 MeV (2) 23.6 MeV
(3) 26.3 eV (4) 23.6 eV

22. If the electric field component of an electromagnetic wave moving in positive *x*-direction is given by $\vec{E} = 6\cos\left[1.2x - 3.6\times10^{8}t\right]\hat{j}\ \frac{N}{C}$, then the average energy density of the electromagnetic wave will be

(1) 2.8×10^{-10} J/m^3 (2) 1.6×10^{-10} J/m^3
(3) 9×10^{10} J/m^3 (4) 3.8×10^{-8} J/m^3

23. **Assertion (A) :** In glass, orange light travels faster than blue light.
Reason (R) : Orange light has a wavelength greater than blue light.

(1) Both (A)and (R) are true and (R) is the correct explanation of (A)
(2) Both (A)and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

24. A ray of light passes through a medium whose refractive index varies with distance as $\mu = \mu_0\left(1+\frac{x}{a}\right)$. If ray enters the medium parallel to *x*-axis, the time taken for ray to travel between *x* = 0 to *x* = a is

(1) $\frac{\mu_0 a}{2C}$ (2) $\frac{3\mu_0 a}{2C}$
(3) $\frac{\mu_0 a}{C}$ (4) $\frac{2\mu_0 a}{3C}$

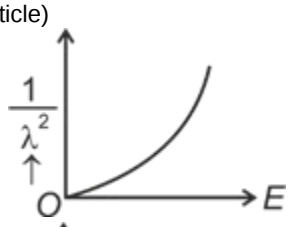
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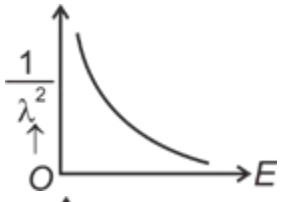
25. A screw gauge has a least count of 0.5 mm and its circular scale is divided into 100 equal divisions. The pitch of screw gauge is
(1) 0.5 mm (2) 0.5 cm
(3) 5 cm (4) 0.05 mm

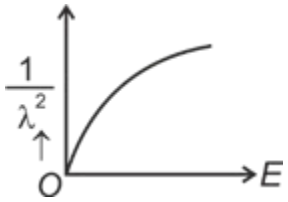
26. In the nuclear decay given below
$_Z^AX \rightarrow {}_{Z+1}{}^AY \rightarrow {}_{Z-1}^{A-4}B* \rightarrow {}_{Z-1}^{A-4}B$
the particles emitted in the sequence are
(1) β^-, α, γ
(2) γ, β^-, α
(3) α, β^-, γ
(4) β^-, γ, α

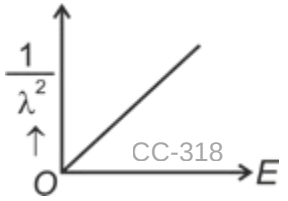
27. In Young's double slit experiment, the distance between two slits is halved and the distance between the screen and slit is made three times. Then width of the fringe becomes
(1) Double (2) Four times
(3) Six times (4) Half

28. The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, *E* is (where λ is de Broglie wavelength of a free particle)

(1) 

(2) 

(3) 

(4) 

29. The electric field in a plane electromagnetic wave is given by, $E_y = 90\ \hat{j}\cos\left(10x + 3\times10^9 t\right)$ V/m. The correct expression for the corresponding magnetic field is (here subscripts denote the direction of the field) [Here *x* is in metre and *t* is in second]
(1) $B_z = 2\times10^{-7}\left(-\hat{k}\right)\cos\left(10x+3\times10^9 t\right)$
(2) $B_z = 3\times10^{-7}\left(-\hat{k}\right)\cos\left(10x+3\times10^9 t\right)$
(3) $B_y = 2\times10^{-7}\left(-\hat{j}\right)\cos\left(10x+3\times10^9 t\right)$
(4) $B_y = 3\times10^{-7}\left(\hat{j}\right)\cos\left(10x+3\times10^9 t\right)$

30. Two polarizers are crossed so that no light is transmitted when only these two are present. A third polarizer is now placed between them such that its polarization axis makes an angle of 30° with the axis of the first polarizer. If the intensity after the first polarizer is I_0, then the intensity of light emerging from the final polarizer will be
(1) $\frac{I_0}{16}$ (2) $\frac{3I_0}{16}$
(3) $\frac{I_0}{8}$ (4) $\frac{I_0}{4}$

31. The mean ratio of first and third Bohr's radius of hydrogen atom is
(1) 1 : 3 (2) 1 : 2
(3) 1 : 6 (4) 1 : 9

32. If the value of Rydberg constant (*R*) is 10^7 m^{-1}, then wave number of the third line of Paschen series in hydrogen spectrum will be
(1) 6.25×10^5 m^{-1} (2) 5.33×10^5 m^{-1}
(3) 7.11×10^5 m^{-1} (4) 8.33×10^5 m^{-1}

33. Which of the following phenomenon is used in optical fibre?
(1) Refraction (2) Scattering
(3) Diffraction (4) Total internal reflection

34. In YDSE, slits are separated by 0.6 mm and screen is placed $2\,m$ away. A beam of light having two wavelengths 520 nm and 620 nm is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is nearly
(1) 102 mm (2) 54 mm
(3) 36 mm (4) 48 mm

35. The total energy of an electron in ground state of hydrogen atom is –13.6 eV. The kinetic energy of an electron in ground state of Li^{2+} ion will be
(1) 122.4 eV (2) –122.4 eV
(3) 244.8 eV (4) –244.8 eV

36. Match the column-I with column-II

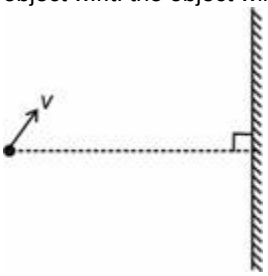
	Column-I		Column-II
(i)	Bohr radius	(p)	$\propto \frac{1}{n}$
(ii)	Velocity	(q)	$\propto \frac{1}{n^2}$
(iii)	Energy	(r)	$\propto n$
(iv)	Orbit angular momentum	(s)	$\propto n^2$

(1) (i) → (p); (ii) → (s); (iii) → (r); (iv) → (q)
(2) (i) → (r); (ii) → (p); (iii) → (s); (iv) → (q)
(3) (i) → (s); (ii) → (p); (iii) → (q); (iv) → (r)
(4) (i) → (p); (ii) → (s); (iii) → (q); (iv) → (r)

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37. Two coherent monochromatic light beams of intensities *I* and 4*I* are superposed. The ratio of maximum and minimum possible intensities in the resulting beam is
(1) 2 : 1 (2) 5 : 3
(3) 9 : 1 (4) 4 : 1

38. An object is moving with velocity $\vec{v} = \left(4\hat{i} + 6\hat{j}\right)$m/s in front of plane mirror as shown in figure. The velocity of image of object w.r.t. the object will be



(1) $\left(-12\hat{j}\right)$ m/s (2) $\left(-8\hat{i}\right)$ m/s
(3) $\left(-4\hat{i}-6\hat{j}\right)$ m/s (4) $\left(-8\hat{i}-12\hat{j}\right)$ m/s

39. In an experiment with vernier callipers of a least count 0.01 cm, when two jaws are joined together the zero of vernier scale lies right to the zero of main scale and 5th division of vernier scale coincides with main scale division. The magnitude of zero error in the measurement is
(1) 0.05 cm (2) 0.10 cm
(3) 0.005 cm (4) Zero

40. The threshold frequency of a photelectric metal is ν_0. If light of frequency $4\nu_0$ is incident on this metal, then the maximum kinetic energy of emitted electrons will be :
(1) $4h\nu_0$ (2) $h\nu_0$
(3) $2h\nu_0$ (4) $3h\nu_0$

41. The ionization potential of hydrogen atom is 13.6 V. An electron in the ground state of hydrogen atom absorbs a photon of energy 12.75 eV. How many different spectral lines can be expected in emission spectrum?
(1) 1 (2) 4
(3) 2 (4) 6

42. In a compound microscope, focal length of objective and eyepiece are 2 cm and 5 cm respectively. An object is placed at a distance 2.2 cm from objective. If final image is formed at least distance of distinct vision (25 cm), then magnifying power of microscope and approximate distance between objective and eyepiece will be respectively
(1) 60, 18 cm (2) 60, 26 cm
(3) 50, 20 cm (4) 50, 26 cm

43. In Young's double slit experiment, fourth bright fringe is formed at a distance *x* from central fringe. The distance between it and second dark fringe on the same side of central maxima is
(1) $\frac{3x}{8}$ (2) $\frac{11x}{8}$
(3) $\frac{13x}{8}$ (4) $\frac{5x}{8}$

44. An EM wave propagates through a non-conducting medium with relative permittivity ε_r = 9 and relative permeability μ_r = 4. The electric field amplitude is E_0 = 90 V/m. Find the amplitude of the magnetic field.
(1) 3×10^{-7} T (2) 5.8×10^{-7} T
(3) 1×10^{-7} T (4) 1.8×10^{-6} T

45. An electron in an atom makes transition from the energy state, $E_1 = -\frac{3}{2}E$ to $E_2 = -3E$, then the wavelength of the emitted photon is
(1) $\frac{hc}{3E}$ (2) $\frac{2hc}{3E}$
(3) $\frac{3hc}{2E}$ (4) $\frac{3hc}{E}$

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CHEMISTRY

46. The osmotic pressure of 5% (mass/volume) solution of cane sugar at 50°C is
(1) 2.51 atm (2) 3 atm
(3) 3.88 atm (4) 3.45 atm

47. The given table has values of Henry's law constant for some gases in water at 293 K.

Gas	K_H (Kbar)
He	144.97
H_2	69.16
N_2	76.48
O_2	34.86

The correct order of solubility of given gases in water is
(1) He < H_2 < N_2 < O_2 (2) He < N_2 < H_2 < O_2
(3) O_2 < N_2 < H_2 < He (4) N_2 < O_2 < H_2 < He

48. 10 g of a non-electrolyte solute is dissolved in 114 g of n-octane, it reduces vapour pressure to 90%. The molar mass (in g mol^{-1}) of the solute is
(1) 45 (2) 60
(3) 90 (4) 80

49. 5 moles of liquid X and 10 moles of liquid Y are mixed to form a solution having vapour pressure of 75 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Based on the given information consider the following statements:
(a) The solution shows positive deviation from Raoult's law.
(b) The A–B interaction is greater than A–A and B–B interaction.
(c) Heat is absorbed during mixing.
The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

50. Largest freezing point depression, is shown by
(1) 0.1 m $K_4[Fe(CN)_6]$ (2) 0.2 m $C_6H_{12}O_6$
(3) 0.1 m $Al_2(SO_4)_3$ (4) 0.3 m KCl

51. Choose the incorrect statement among the following:
(1) $\mathrm{Pt, H_2|HCl(C_1)||HCl(C_2)|Pt, H_2}$ represents a cell with $\mathrm{E^o_{cell} = 0}$
(2) On increasing the dilution, specific conductance decreases
(3) On increasing the dilution, molar conductivity decreases
(4) O_2(g) is produced at anode when an aqueous solution of sodium sulphate is electrolysed using inert electrodes

52. Consider the given four electrolytes and the value of their respective conductivities

A. 2×10^{-5} ohm^{-1} cm^{-1}

B. 1×10^{-8} ohm^{-1} cm^{-1}

C. 5×10^{-6} ohm^{-1} cm^{-1}

D. 9×10^{-2} ohm^{-1} cm^{-1}

The correct order of the resistance offered by them for the passage of electric current in the same electrolytic cell is

(1) A > B > C > D (2) A > D > C > B

(3) B > D > A > C (4) B > C > A > D

53. Identify the incorrect match with respect to the product obtained during electrolysis of the given electrolytes using inert electrode.

(1) Molten lead bromide – Br_2 at anode

(2) Aqueous sodium chloride – H_2 at cathode

(3) Aqueous sodium chloride – O_2 at anode

(4) Dilute H_2SO_4 – O_2 at anode

54. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 5 and by passing hydrogen gas around the platinum wire at one atm pressure at 298 K temperature. The reduction potential of the electrode would be

(1) –0.295 V (2) –0.148 V

(3) +0.295 V (4) +0.148 V

55. The number of electron flown through a metallic wire if a steady current of 1 ampere is passed through it for 2 hrs, is

(1) 1.25×10^{18} (2) 4.5×10^{22}

(3) 6.25×10^{21} (4) 7.5×10^{19}

56. Correct expression of E°_{cell} with equilibrium constant is

(1) $E^{\circ}_{cell} = \log K_c$ (2) $E^{\circ}_{cell} = \frac{2.303\ nF}{RT} \log K_c$

(3) $E^{\circ}_{cell} = \frac{2.303\ RT}{nF} \log K_c$ (4) $E^{\circ}_{cell} = \frac{nF}{RT}$

57. The unit of rate of reaction for a second order reaction is

(1) mol^{-1} L s^{-1} (2) mol L^{-1} s^{-1}

(3) mol^{2} L^{-2} s^{-1} (4) mol^{-2} L^{2} s^{-1}

58. For a first order reaction $A \rightarrow$ Products, initial concentration of A is $0.2M$ which becomes $0.0125M$ after 50 minutes. The rate constant of the reaction is

(1) 9.2 x 10^{-4} s^{-1} (2) 2.4 x 10^{-3} s^{-1}

(3) 7.5 x 10^{-3} s^{-1} (4) 1.6 x 10^{-4} s^{-1}

59. Incorrect statement among the following is

(1) Unit of rate constant for second order reaction is mol^{-1} L s^{-1}

(2) Molecularity of a reaction is applicable only for elementary reactions.

(3) Inversion of cane sugar in acidic medium is an example of pseudo first order reaction.

(4) Thermal decomposition of HI on gold surface is an example of first order reaction.

60. A drop of a solution having a volume of 0.4 ml contains 5×10^{-6} moles of $OH^{\ominus}$. If the rate constant for disappearance of OH^- is 10^6 mol L^{-1} s^{-1}. How long would it take for $\overset{\ominus}{O}H$ in drop to disappear?

(1) 1.25×10^{-6} second (2) 1.25×10^{-8} second

(3) 1.25×10^{-9} second (4) 1.25×10^{-5} second

61. Consider the following statements regarding to the equation, Rate = $PZ_{AB}e^{-E_a/RT}$

(a) Z_{AB} represents the collision frequency of reactants, A and B.

(b) $e^{-E_a/RT}$ represents the fraction of molecules with energies less than E_a.

(c) P takes into account the fact that in a collision, molecules must be properly oriented.

Select the correct statements.

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

62. Aqueous solution of which of the following complexes will consume maximum equivalents of $AgNO_3$?

(1) $[Co(NH_3)_4Cl_2]Cl$ (2) $[Co(NH_3)_6]Cl_3$

(3) $[Co(NH_3)_5Cl]Cl_2$ (4) $[Co(NH_3)_3Cl_3]$

63. Fusion of chromite ore with sodium carbonate in free access of air results in compound (X), Fe_2O_3 and a gaseous product CO_2. (X) is acidified with H_2SO_4 to give a coloured compound (Y). The products (X) and (Y) respectively are

(1) $NaCrO_2$ and Na_2CrO_4 (2) Na_2CrO_4 and $Na_2Cr_2O_7$

(3) $Na_2Cr_2O_7$ and Na_2CrO_4 (4) $NaCrO_2$ and $Na_2Cr_2O_7$

64. Consider the following statements:

(a) Noble gases are sparingly soluble in water

(b) Positive electron gain enthalpy of argon is lesser than helium

(c) Helium is a non-inflammable and light gas

The correct statement(s) is/are

(1) (b) and (c) only (2) (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

65. Consider the following statements:

(a) Monoclinic sulphur is soluble in CS_2 but rhombic sulphur is not

(b) Both rhombic and monoclinic sulphur have S_8 molecules

(c) Rhombic sulphur is stable below 369 K and transforms into monoclinic sulphur above this

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

66. Consider the following statements:

(a) N_2O is a colourless gas

(b) N_2O can be obtained by heating NH_4NO_3

(c) N_2O_3 is a blue solid and is acidic in nature

(d) N_2O_3 can be obtained by heating lead nitrate at 673 K.

The correct statements are

(1) (a) and (b) only (2) (a), (b) and (c) only

(3) (a), (c) and (d) only (4) (a), (b), (c) and (d)

67. Which of the following species are diamagnetic?

(a) $[Ni(CO)_4]$

(b) $[ZnCl_4]^{2-}$

(c) $[Fe(CN)_6]^{3-}$

(d) $[FeCl_6]^{4-}$

(e) $[TiCl_3(H_2O)_3]$

Choose the correct answer from the options given below:

(1) (a) and (b) only (2) (b), (c) and (e) only

(3) (a), (c) and (d) only (4) (a), (b) and (e) only

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68. Consider the following statements
(a) The single N – N bond is weaker than the single P – P bond.
(b) Phosphorus and arsenic can form $d\pi - d\pi$ bond with transition metals when their compounds like $P(C_2H_5)_3$ and $As(C_6H_5)_3$ act as ligands.
(c) Thermal decomposition of ammonium dichromate gives ammonia gas as one of the products.
The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a), (b) and (c) (4) (a) and (c) only

69. Number of mole(s) of permanganate ions required to react completely with 2 moles of thiosulphate ions in faintly alkaline medium is
(1) 4 (2) $\frac{8}{3}$
(3) $\frac{16}{3}$ (4) $\frac{4}{5}$

70. Correct order of ionic radii of the given species is
(1) $Nd^{3+} > Dy^{3+} > Eu^{3+} > Yb^{3+}$
(2) $Eu^{3+} > Nd^{3+} > Dy^{3+} > Yb^{3+}$
(3) $Yb^{3+} > Dy^{3+} > Eu^{3+} > Nd^{3+}$
(4) $Nd^{3+} > Eu^{3+} > Dy^{3+} > Yb^{3+}$

71. The organometallic compound which will react with sodium metal most easily is
(1) $Ni(CO)_4$ (2) $Fe(CO)_5$
(3) $V(CO)_6$ (4) $Cr(CO)_6$

72. Complex showing geometrical isomerism is
(1) $[Fe(H_2O)_6]Cl_3$ (2) $[Co(NH_3)_4Cl_2]Cl$
(3) $K_2[PtCl_4]$ (4) $[Co(en)_3]Cl_3$

73. The correct order of covalent nature is
(1) $NF_3 > PF_3 > BiF_3 > SbF_3$
(2) $NF_3 > BiF_3 > PF_3 > SbF_3$
(3) $NF_3 > PF_3 > SbF_3 > BiF_3$
(4) $NF_3 > SbF_3 > BiF_3 > PF_3$

74. Ce (Z = 58) and Yb (Z = 70) exhibit stable +4 and +2 oxidation state respectively. This is because
(1) Ce^{4+} and Yb^{2+} acquire f^7 configuration
(2) Ce^{4+} and Yb^{2+} acquire f^0 configuration
(3) Ce^{4+} and Yb^{2+} acquire f^0 and f^{14} configuration
(4) Ce^{4+} and Yb^{2+} acquire f^7 and f^{14} configuration

75. Chlorine is manufactured by Deacon's process. Chemical equation involved in this process is
(1) $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$
(2) $4NaCl + MnO_2 + 4H_2SO_4 \rightarrow MnCl_2 + 4NaHSO_4 + 2H_2O + Cl_2$
(3) $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$
(4) $4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$

76. Acidic oxide among the following is
(1) Mn_2O_7 (2) ZnO
(3) CrO (4) Al_2O_3

77. Match the given compounds in column I with the hybridisation and shape given in column II.

Column I	Column II
(i) XeF_4	(A) sp^3d^2 – Square pyramidal
(ii) $XeOF_4$	(B) sp^3d – See saw
(iii) XeO_2F_2	(C) sp^3 – Pyramidal
(iv) XeO_3	(D) sp^3d^2 – Square planar

(1) (i) → D , (ii) → B , (iii) → A , (iv) → C
(2) (i) → D , (ii) →A , (iii) → B , (iv) → C
(3) (i) → A , (ii) → D , (iii) → B , (iv) → C
(4) (i) → D , (ii) → C , (iii) → B , (iv) → A

78. Consider the following statements about interstitial compounds
(a) They are highly reactive.
(b) They do not retain metallic conductivity.
(c) They have high melting points, higher than those of pure metals.
The incorrect statement(s) is/are
(1) (a) and (b) only (2) (c) only
(3) (a), (b) and (c) (4) (b) and (c) only

79. Excess of copper and iron are removed mainly by which of the following chelating ligands via formation of coordination compounds?
(1) Desferrioxime B and EDTA respectively
(2) EDTA and D-Penicillamine respectively
(3) Desferrioxime B and D-Penicillamine respectively
(4) D-Penicillamine and Desferrioxime B respectively

80. In which of the following oxoacids of phosphorous, number of P-OH bonds is maximum?
(1) Pyrophosphorous acid (2) Hypophosphorous acid
(3) Hypophosphoric acid (4) Orthophosphoric acid

81. Match list-I with list-II.

	List-I Species		List-II Crystal Field Stabilization Energy (CFSE) [Ignoring pairing energy]
a.	$[Mn(H_2O)_6]^{2+}$	(i)	$-0.4\Delta_0$
b.	$[Fe(CN)_6]^{4-}$	(ii)	Zero
c.	$[Cr(H_2O)_6]^{3+}$	(iii)	$-2.4\Delta_0$
d.	$[Fe(H_2O)_6]^{2+}$	(iv)	$-1.2\Delta_0$

The correct match is
(1) a(ii), b(iii), c(i), d(iv) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(ii), c(iv), d(i) (4) a(iii), b(ii), c(i), d(iv)

82. Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is
(1) $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(H_2O)_6]^{3+}$
(2) $[Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+}$
(3) $[Co(H_2O)_6]^{3+} < [Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+}$
(4) $[Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+} < [Co(H_2O)_6]^{3+}$

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83. Which of the following statement(s) is/are correct?
(A) H_2S gas turns lead acetate paper black
(B) Methanamine and N-Methylmethanamine can be distinguished by Hinsberg's reagent
(C) The colour of phenolphthalein indicator is pink in basic medium
Choose the correct option.
(1) (A) and (B) only (2) (B) and (C) only
(3) (A) and (C) only (4) (A), (B) and (C)

84. Given below are two statements:
Statement I: If enthalpy of neutralisation of acetic acid by NaOH is –x kJ/mol, then enthalpy of ionisation of acetic acid is (57.1 – x) kJ/mol.
Statement II: Lyophilic sols are stabilized due to solvation of colloidal particles by dispersion medium while stability of lyophobic sol is due to charge on colloidal particles.
In the light of above statements, choose the correct answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

85. Identify the correct group reagent for identification of Cu^{2+} ion
(1) H_2S gas in presence of dil. HCl
(2) NH_4OH in presence of NH_4Cl
(3) H_2S in presence of NH_4OH
(4) $(NH_4)_2CO_3$ in presence of NH_4OH

86. Which of the following set of alcohols give fastest reaction with Lucas reagent?
(1) Isopropyl alcohol and neopentyl alcohol
(2) Neopentyl alcohol and isopentyl alcohol
(3) tert-Butyl alcohol and 2-Methylbutan-2-ol
(4) tert-Butyl alcohol and 3-Methylbutan-2-ol

87. Given below are two statements:
Statement I: On addition of β-naphthol to benzene diazonium salt, coupling reaction takes place to form a scarlet red coloured dye.
Statement II: Phenols form violet coloured complex with freshly prepared $FeCl_3$ solution.
In the light of above statements, choose the correct answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

88. Colour of $[Fe(SCN)]^{2+}$ is
(1) Yellow (2) Blood red
(3) Orange (4) Deep blue

89. Alcoholic compounds on reaction with ceric ammonium nitrate gives a red colouration due to the formation of
(1) $[Ce(NO_3)_4\ (ROH)_3]$
(2) $[Ce(NH_4)_4(ROH)_3](NO_3)_3$
(3) $[Ce(H_2O)_4(ROH)_2](NO_3)_2$
(4) $[Ce(ROH)_6](NO_3)_2$

90. Consider the following statements
Statement I : Acetylation of aniline is carried out in the presence of weaker base than aniline.
Statement II : Aniline is weaker base than pyridine.
Choose the correct option
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

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BOTANY

91. Select the **incorrect** match w.r.t. ploidy level of given structures in angiosperms.
(1) Nucellus – 2 n
(2) Megaspore mother cell – 2 n
(3) Functional megaspore – n
(4) Female gametophyte – 2 n

92. Monoecious plants can prevent
(1) Both autogamy and geitonogamy
(2) Autogamy but not geitonogamy
(3) Geitonogamy
(4) Neither autogamy nor geitonogamy

93. What is the fate of antipodal cell after double fertilisation?
(1) It forms the envelope of embryo
(2) It degenerates
(3) It divides and forms the part of endosperm
(4) It forms the cotyledon of the seed

94. **Statement A:** The distal end of the filament of stamen are attached to the thalamus or petal of the flower.
Statement B: In typical angiosperm, anther is bilobed with each lobe having four theca.
In the light of above statements, choose the **correct** answer from the options given below
(1) Statement A is correct but statement B is incorrect.
(2) Statement A is incorrect but statement B is correct.
(3) Both statement A and statement B are correct
(4) Both statement A and statement B are incorrect

95. Vegetative cell of male gametophyte in flowering plants
(1) Is smaller than the generative cell
(2) Has large irregularly shaped nucleus
(3) Floats in the cytoplasm of generative cell
(4) Has dense cytoplasm and is spindle shaped

96. Egg apparatus of embryo sac contains
(1) Two egg cells & one synergid
(2) One egg cell & two synergids
(3) One egg cell & two antipodal cells
(4) One egg cell & one synergid only

97. Site of syngamy and triple fusion in the flowering plant is
(1) Chalaza (2) Anther
(3) Endosperm (4) Embryo sac

98. Read the following Assertion **(A)** and Reason **(R)** statements:
Assertion (A): In angiosperms, seed formation is more dependable.
Reason (R): In angiosperms, reproductive processes such as pollination and fertilisation are independent of water.
In the light of the above statements, select the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

99. An event unique to flowering plants is
(1) Embryo formation (2) Seed formation
(3) Syngamy (4) Triple fusion

100. Read the following statements w.r.t double fertilisation.
a. PEN develops into endosperm
b. One male gamete fuse with haploid secondary nucleus
c. It is an event unique to gymnosperms & angiosperms
d. One of the male gamete fuses with the diploid secondary nucleus to produce PEN
Mark the option representing **incorrect** statements
(1) a, d (2) a, b
(3) c, d (4) b, c

101. Read the following statements are select the **correct** option
Statement A : Parthenocarpic fruit are seedless and can be produced through application of growth hormones.
Statement B : Seeds form the basis of our agriculture as they show dehydration and dormancy.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are incorrect
(4) Both the statements A and B are correct

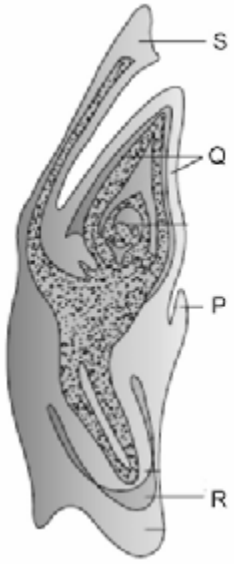
102. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Megaspore mother cell	(i)	One of the diploid cells undergoes dedifferentiation to form parent cell of female gametophyte
b.	Nucellus	(ii)	It undergoes equational division to form the female gametophyte
c.	Embryo sac	(iii)	It undergoes reductional division to form haploid cells
d.	Functional megaspore	(iv)	It is formed after three sequential equational divisions that is strictly free nuclear, after that cell walls are laid down

(1) a(iii), b(iv), c(i), d(ii) (2) a(iii), b(i), c(iv), d(ii)
(3) a(iv), b(ii), c(i), d(iii) (4) a(iii), b(ii), c(i), d(iv)

103. Select the characteristic features that are favourable for pollination in anemophilous flowers.
a. Flowers packed into inflorescence
b. Presence of nectaries
c. Well exposed stamens
d. Light and sticky pollens
e. Feathery stigma
f. Foul odour
Choose the option for **correct** ones.
(1) a, b, d and f (2) c and e only
(3) a, c and e (4) a, b, e and f

104. Observe the following diagram.



Based on the above given figure, match the labels (Column-I) with their **correct** description (Column-II).

Column - I		Column - II	
I.	P	a.	Covers the terminal structure of hypocotyl
II.	Q	b.	Cotyledon situated laterally to the embryonal axis
III.	R	c.	Remains of second cotyledon
IV.	S	d.	Hollow foliar structure enclosing shoot apex and few leaf primordia

(1) I-c, II-d, III-a, IV-b (2) I-a, II-d, III-c, IV-b
(3) I-b, II-a, III-d, IV-c (4) I-c, II-b, III-a, IV-d

105. Choose the **incorrect** match..
(1) Chasmogamous flower – *Commelina*
(2) Fresh water plant – *Hydrilla*
(3) Wind pollinated plant – Maize
(4) Water pollinated – *Eichhornia*

106. In situ conservation strategy is
(1) Wild life safari parks (2) Botanical gardens
(3) Zoological parks (4) Wild life sanctuaries

107. Loss of biodiversity in a region may lead to
(1) Greater resistance to drought
(2) Decline in plant production
(3) Decreased variability in certain ecosystem processes
(4) Decreased pest and disease cycles

108. Initially, how many biodiversity hot spots were identified globally?
(1) 25 (2) 34
(3) 20 (4) 30

109. Maximum biodiversity is of
(1) Insects in vertebrates, fishes in invertebrates, and lichens in plants
(2) Molluscs in invertebrates, amphibians in vertebrates and mosses in plants
(3) Insects in invertebrates, fishes in vertebrates
(4) Crustaceans in vertebrates, birds in invertebrates and algae in plants

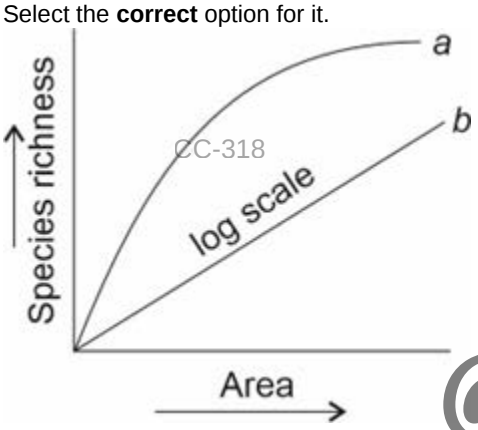
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110. Hot spots are characterised by
(i) Large number of endemic species
(ii) Low levels of species richness
(iii) Regions of accelerated habitat loss
(iv) Exceptionally high biodiversity regions
The **correct** ones are
(1) (i), (ii) & (iii) (2) (i), (iii) & (iv)
(3) (ii), (iii) & (iv) (4) (i), (ii) & (iv)

111. Presently how much percentage of mammal species are facing the threat of extinction?
(1) 12% (2) 23%
(3) 31% (4) 32%

112. Select the **incorrect** match w.r.t sacred groves and their locations.
(1) Khasi and Jaintia Hills – Meghalaya
(2) Aravali Hills – Rajasthan
(3) Western ghats – Maharashtra
(4) Bastar – Karnataka

113. The following graph shows the species area relationship. Select the **correct** option for it.



(1) For large areas Z lies in the range of 0.1 to 0.2 regardless of the taxonomic group
(2) The value of Z lies between 0.6 to 1.2 for an entire continent
(3) The slope of line b becomes much steeper for small areas
(4) If the value of Z is more the slope will be less steeper

114. Mark true **(T)** or false **(F)** for the following statements and select the **correct** option.
A. Term biodiversity was popularised by a sociobiologist to describe the combined heterogeneity at all levels of biological organisation.
B. More than 50,000 genetically different strains of rice occur in India.
C. Western Ghats have greater amphibian species diversity as compared to Eastern Ghats.
D. India has only 8.1 Per cent of the world's land area, its share of the global speces diversity is only 2.4 per cent.
(1) A-(T), B-(F), C-(F), D-(F) (2) A-(F), B-(F), C-(T), D-(F)
(3) A-(T), B-(T), C-(T), D-(F) (4) A-(T), B-(T), C-(F), D-(F)

115. Match the following column I with column II and choose the **correct** option

Column-I (Animal)	**Column-II (Place of extinction)**
A. Dodo	(i) Australia
B. Stellar's sea cow	(ii) Africa
C. Thylacine	(iii) Mauritius
D. Quagga	(iv) Russia

(1) A-(ii), B-(iii), C-(iv), D-(i) (2) A-(iii), B-(iv), C-(i), D-(ii)
(3) A-(iii), B-(i), C-(iv), D-(ii) (4) A-(iv), B-(i), C-(ii), D-(iii)

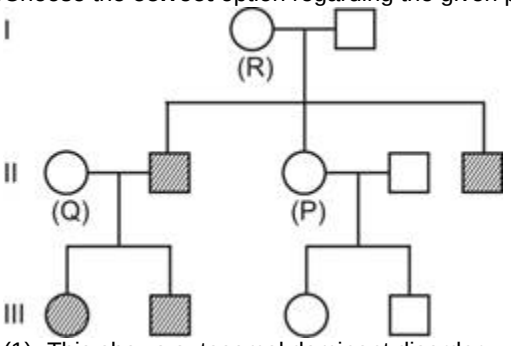
116. Select the **incorrect** match w.r.t. the invasion of alien species and the affected organisms.
(1) *Eichhornia* –Several terrestrial plants
(2) Nile perch – Cichlid fish
(3) *Lantana* – Several plant species in forests
(4) African catfish – Catfishes in our rivers

117. Read the following statements
(a) Too much variation in productivity from year to year.
(b) Resilience to occasional natural or man-made disturbances.
(c) Resistance to invasion of alien species.
Which of the above mentioned statements is/are **correct** for a stable community?
(1) Only (a) (2) Only (b) and (c)
(3) Only (c) (4) Only (a) and (b)

118. The historical 'Convention on Biological Diversity' known as 'Earth Summit' was held in
(1) South Africa in 1999 (2) Montreal in 1987
(3) Rio-de-Janeiro in 1992 (4) Johannesburg in 1992

119. Experimental verification of the 'chromosomal theory of inheritance' was done by
(1) T.H. Morgan (2) Tschermak
(3) Hugo de Vries (4) Correns

120. Choose the **correct** option regarding the given pedigree



(1) This shows autosomal dominant disorder
(2) The genotype of II (Q) is X^CX
(3) The I(R) female is homozygous for the disorder
(4) The genotype of II(P) is X^CX^C

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121. Read the following statements based on the Mendelian observations and select the **correct** option.
P: A recessive character that was not expressed in heterozygous condition may be expressed again when it becomes homozygous.
Q: The expression of the characters in the offsprings, follow a definite pattern in different generations-first (F_1), second (F_2) and so on.
R: The recessive characters are only expressed in homozygous conditions.
S: Characters segregate during formation of gametes.
(1) P and Q are true and Q is the reason for P
(2) R and Q are true but P and S are false
(3) All the given statements are true
(4) Q and S are true and Q is the reason for S

122. Read the following statements and find out the **correct** ones
(a) Occasionally a single gene product may produce more than one effect.
(b) Dominance is an autonomous feature of a gene.
(c) Frequency of recombination does not exceed 50% when two genes assort independently.
(d) F_2 dihybrid test cross ratio is 1 : 1 : 1 : 1 when genes are linked completely.
(1) (a) & (c) (2) (a) & (d)
(3) (b) & (c) (4) (b) & (d)

123. A pea plant, heterozygous for both seed shape and seed colour was self crossed. Out of 480 seeds obtained by this cross, total number of wrinkled-green seeds will be
(1) 150 (2) 40
(3) 30 (4) 248

124. The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome?
(1) a, b, c, d (2) a, c, b, d
(3) a, d, b, c (4) d, b, a, c

125. Identify the following statements as true **(T)** or false **(F)** and choose the option accordingly.
(A) The law of dominance is used to explain the expression of only one of the parental characters in the F_1 generation in a monohybrid cross.
(B) Factors always occur in pairs whether the organism is haploid or diploid.
(C) Law of segregation is based on the fact that two characters located on different chromosomes get segregated during gamete formation.
(D) In incomplete dominance, the individuals in F_1 generation had a phenotype that does not resemble either of the two parents and is in between the two.

	(A)	(B)	(C)	(D)
(1)	F	F	T	T
(2)	T	F	F	T
(3)	T	T	F	F
(4)	F	T	T	F

(1) (1) (2) (2)
(3) (3) (4) (4)

126. *Drosophila melanogaster* is suitable for genetic studies because
(1) It is unable to grow on synthetic medium
(2) It can produce large number of progenies in a single mating
(3) It completes its life cycle in only two years
(4) Its male and female individuals are not easily distinguishable

127. Select the organism in which egg determines the sex of the offspring?
(1) Grasshopper (2) *Drosophila*
(3) Human (4) Bird

128. An individual affected with Turner's syndrome
(1) Lacks X – chromosomes
(2) Has an additional copy of X – chromosome
(3) Is monosomic due to the presence of only one X – chromosome
(4) Has trisomic condition of an autosome

129. Select the set of parents that cannot produce child with blood group 'O'.
(a) $I^A i \times I^A i$
(b) $I^B i \times I^A I^A$
(c) $ii \times I^A i$
(d) $I^B I^B \times I^A I^B$
(1) (b) & (d) (2) (a) & (d)
(3) (b) & (c) (4) All except (d)

130. Select the syndrome or disorder resulting due to non-disjunction of sex-chromosome
(1) Down's syndrome (2) Edward's syndrome
(3) Klinefelter's syndrome (4) Phenylketonuria

131. Which of the following character(s) is/are not related to Down's syndrome?
(1) Short stature and small round head
(2) Furrowed tongue and partially open mouth
(3) Big and wrinkled tongue
(4) Gynaecomastia

132. Choose the **incorrect** one for flower colour inheritance in *Antirrhinum.*
(1) Plants with RR genotype produces red flowers
(2) It shows incomplete dominance
(3) When F_1 plants (Rr) are self pollinated, the genotype and phenotype ratio are same in F_2 generation
(4) The genotype rr produces pink coloured flowers

133. In a typical Mendelian monohybrid cross, a cross between F_1 hybrid and a dominant parent gives a genotypic ratio of
(1) 1 : 1 (2) 1 : 2
(3) 3 : 1 (4) 1 : 4

134. Select the **incorrect** match.
(1) Starch grain size in pea – incomplete dominance
(2) Seed shape in pea – Complete dominance
(3) ABO blood group – Multiple allelism
(4) Flower colour in dog flower – Co-dominance

135. A normal visioned man whose father was colourblind, marries a woman whose mother was colourblind but father was normal. They had their first child as son. What are the chances that this child would be colourblind?
(1) 25% (2) 50%
(3) 75% (4) 100%

ZOOLOGY

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136. Which contraceptive method delays conception for longest period when applied or taken only once?
(1) Saheli pill (2) Diaphragm
(3) Implant (4) Cervical cap

137. Darwin's finches represent
(1) Adaptive divergence (2) Adaptive convergence
(3) Analogy (4) Artificial selection

138. Select the correct option w.r.t. techniques that involve use of X-rays for cancer detection
a. Computed tomography
b. MRI
c. Biopsy
d. Radiography
e. Immunotherapy
(1) b and c (2) b, d and e
(3) a and d (4) a only

139. Choose the **odd** one w.r.t. copper releasing IUDs.
(1) CuT (2) Progestasert
(3) Cu7 (4) Multiload 375

140. Choose the **incorrect** option w.r.t. primitive atmosphere on earth.
(1) Reducing atmosphere
(2) CH_4, NH_3, H_2 and water vapour were present
(3) High temperature and volcanic eruptions
(4) Presence of ozone to prevent entry of UV rays into the atmosphere

141. Transfer of early embryo with upto eight blastomeres into the fallopian tube is called
(1) GIFT (2) AI
(3) ZIFT (4) IUI

142. How many of the following are ideal contraceptives that can alter/interfere with the sexual drive/act of the user?
Condom, Injectables, IUDs, Oral pills, Saheli
Select the **correct** option.
(1) One (2) Two
(3) Three (4) Zero

143. Select the **odd** pair w.r.t. convergent evolution.
(1) Bobcat and Tasmanian tiger cat
(2) Numbat and wombat
(3) Lemur and spotted cuscus
(4) Wolf and Tasmanian wolf

144. When HIV infects a person and enters macrophages, the enzyme which helps in formation of viral DNA from viral RNA is
(1) DNA ligase (2) RNA polymerase
(3) Reverse transcriptase (4) *Taq* polymerase

145. If the frequency of the dominant allele in a population is 0.7, calculate the frequency of heterozygous individuals in the population.
(1) 0.3 (2) 0.21
(3) 0.42 (4) 0.7

146. Presence of which of the following does **not** disrupt the Hardy-Weinberg equilibrium?
(1) Gene flow (2) Genetic drift
(3) Mutation (4) Random mating

147. All of the following are used as a vaccine which provide active immunity, **except**
(1) A preparation of antigenic proteins of pathogen
(2) Inactivated pathogen
(3) A preparation containing antibodies
(4) A preparation containing killed pathogen

148. Read the following statements and choose the **correct** option.
Statement-A : When our body encounters a pathogen for the first time it produces a response called the primary immune response.
Statement-B : Primary immune response is of lower intensity than the anamnestic response.
(1) Both statements are incorrect
(2) Statement A is correct but statement B is incorrect
(3) Statement A is incorrect but statement B is correct
(4) Both statements are correct

149. Sexual reproduction of *Plasmodium* occurs in
(1) Salivary glands of *Anopheles*
(2) Human RBC
(3) Human liver
(4) Gut of female *Anopheles*

150. Choose the **incorrect** match:

	Column I (Disease)		Column II (Caused by)
(1)	Chlamydiasis	–	Bacterium
(2)	Gonorrhoea	–	Bacterium
(3)	Trichomoniasis	–	Fungus
(4)	AIDS	–	Virus

(1) (1) (2) (2)
(3) (3) (4) (4)

151. Genetic drift is not observed in
(1) Small population
(2) Interbreeding large sized population
(3) Inbreeding population
(4) Isolated population

152. Who concluded that existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago?
(1) Charles Darwin (2) Ernst von Baer
(3) Hugo deVries (4) Lamarck

153. Thorn of *Bougainvillea* and tendrils of *Cucurbita* represent all of the following, except
(1) Homology (2) Divergent evolution
(3) Convergent evolution (4) Common ancestry

154. All of the following drugs are used to quickly reduce symptoms of allergy **except**
(1) Anti-histamine (2) Adrenalin
(3) Glucocorticoids (4) Histamine

155. Match Column I and II w.r.t. action of contraceptives.

	Column I		Column II
a.	Jellies	(i)	Prevents implantation
b.	Progestasert	(ii)	Spermicidal
c.	Copper T	(iii)	Suppress sperm motility
		(iv)	Prevent deposition of sperms into vagina

Choose the **correct** option.
(1) a(iv), b(i), c(ii) (2) a(ii), b(i), c(iii)
(3) a(ii), b(iii), c(iv) (4) a(iv), b(i), c(iii)

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156. Read the following statements carefully.
A) Studies suggest that first organisms that invaded land were plants.
B) Saltations are single step large mutations and Darwinian variations are small and directional.
C) Sauropsids were common ancestors of therapsids and thecodonts.
D) *Homo habilis* was the first hominid with brain capacity ranging between 650-800 cc
Choose the **correct** option w.r.t. True (T) and False (F) statements.

	A	B	C	D
(i)	T	F	T	F
(ii)	F	T	F	T
(iii)	T	T	F	T
(iv)	T	T	F	F

(1) (i) (2) (ii)
(3) (iii) (4) (iv)

157. Which of the following is **not** true statement?
(1) Evolution is a stochastic process
(2) Evolution is not a directed process in the sense of determinism
(3) Evolution is always from simple form to complex form
(4) Natural selection in which more individual acquire value other than the mean character value is called directional selection

158. Select the autoimmune disease among the following.
(1) Ascariasis (2) Rheumatoid arthritis
(3) Filariasis (4) Amoebiasis

159. AIDS was first reported in year 'X' and has spread all over the world killing more than 'Y' persons.
Select the correct option for X and Y respectively.

	X	Y
(1)	1781	25 billion
(2)	1891	2.5 billion
(3)	1981	2.5 crore
(4)	1871	0.25 crore

(1) (1) (2) (2)
(3) (3) (4) (4)

160. HIV belongs to the category of
(1) Rhinoviruses (2) Retroviruses
(3) Gemini viruses (4) Paramyxoviruses

161. Read the given statements and chose the correct option.
Statement (A) : The story of evolution of modern man is most interesting and appears to parallel evolution of human brain and language.
Statement (B) : Agriculture came around 10,000 years back and human settlements started.
(1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Only statement (A) is correct
(4) Only statement (B) is correct

162. Around 3 - 4 mya old fossils of man-like bones have possibly been discovered in Ethiopia and Tanzania. They were most likely evolved from
(1) Neanderthal man (2) *Ramapithecus*
(3) *Homo erectus* (4) *Homo habilis*

163. Fossils are mostly preserved in
(1) Sedimentary rocks (2) Igneous rocks
(3) Metamorphic rocks (4) Water

164. **Assertion (A):** Post-industrialisation, the number of melanised moths increased.
Reason (R): White-winged moths were completely wiped off, due to predation.
In the light of above statements, select the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

165. Select the **correct** statement.
(1) Physical barriers include acid in stomach, saliva in mouth and tears from eye
(2) Rhino viruses infect nose and upper respiratory tract causing common cold but do not infect lungs
(3) Ruptured WBCs release hemozoin which is responsible for chill and high fever in malaria
(4) Interleukins are synthesized within B-lymphocytes for production of antigens

166. Random, non-directional changes in gene frequencies within a small populations that occur by chance or accidents are referred to as
(1) Gene flow (2) Genetic drift
(3) Genetic recombination (4) Gene migration

167. MTP is legal in all of the following cases **except**
(1) Risk to the life of the pregnant women
(2) Possibility of serious physical and mental abnormalities after delivery of child
(3) To get rid of unwanted pregnancy due to casual unprotected intercourse
(4) To get rid of unwanted pregnancy when the foetus is found to be female

168. Marijuana and hashish both belong to the category of
(1) Opioids (2) Cannabinoids
(3) Coca alkaloid (4) Sedatives

169. A pregnant woman went to the doctor for her checkup. Doctor observed the appearance of hair on head of the foetus, but doctor also diagnosed a genetic abnormality in the foetus. For MTP in this scenario opinions of how many registered medical practitioners are required?
(1) One (2) Three
(3) Two (4) Four

170. Immune response primarily responsible for rejection of kidney graft is
(1) Cell-mediated immune response
(2) Inflammatory immune response
(3) Humoral immune response
(4) Auto-immune response

171. Biological response modifiers which activate immune system and aid in destroying tumor are
(1) α-interferons (2) β-interferon
(3) γ-interferon (4) Interleukin

172. Receptors for depressant drug 'X' are present in our CNS and gastrointestinal tract. 'X' can be all except
(1) Smack (2) Heroin
(3) Morphine (4) Coke

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173. Select the pathogen mismatched with the symptoms of disease caused by it from the list given below:
(1) *Entamoeba histolytica* : Constipation, abdominal pain.
(2) *Epidermophyton* : Dry scaly lesions on nail.
(3) *Wuchereria bancrofti* : Chronic inflammation of lymphatic vessels of lower limb.
(4) *Haemophilus influenzae* : Blockage of the intestinal passage.

174. Select **incorrect** statement w.r.t. antibodies.
(1) B-lymphocytes produce army of proteins called antibodies in response to pathogens
(2) The primary and secondary immune response is carried out in our body with the help of antibodies
(3) T-lymphocytes activate B-lymphocytes to produce antibodies for humoral immune response
(4) Each antibody is represented as H_2L_2 having four heavy polypeptide chains

175. Cycads and angiosperms arose from a common ancestor called _____ that existed on earth during ____ period.
Choose the option that correctly fills the blanks respectively.
(1) *Zosterophyllum*, triassic
(2) Chlorophyta, silurian
(3) Seed ferns, carboniferous
(4) Herbaceous lycopods, cretaceous

176. The malignant tumors show all of the following features, **except**
(1) Uncontrolled and rapid growth
(2) Damage to surrounding tissues
(3) Presence of contact inhibition
(4) Metastasis

177. Which of the following statements are correct?
(A) Reproductive health refers to total well-being in all aspects of reproduction
(B) Amniocentesis is legally banned for sex determination in India.
(C) "Saheli" -a new oral contraceptive for females was developed in collaboration with ICMR (New Delhi)
(D) Amniocentesis is used to determine genetic disorders and survivability of foetus.
Choose the **most appropriate** answer from the options given below:
(1) (A), (B) and (D) only (2) (A) and (C) only
(3) (B) and (C) only (4) (D) and (C) only

178. 'Panspermia' is favourite idea for some astronomers in context of origin of life. According to this concept
(1) Life was created by God
(2) Life originated from non-living material like mud and straw
(3) Life came on earth as spores from some other planet
(4) Life arose from some pre-existing life

179. In 1938, a fish caught in South Africa happened to be a Coelacanth which was thought to be extinct. These animals called lobefins evolved into the first
(1) Amphibians (2) Jawless fishes
(3) Reptiles (4) Mammals

180. Select an **incorrect** statement w.r.t. the technique shown in figure



(1) It is a sterlisation procedure
(2) It has poor reversibility
(3) It represents vasectomy
(4) It blocks gamete transport

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